

# **Orbital Station Operations Handbook**

Revision 7 - Northwind Orbital Services

## **Chapter 1: Scope and Applicability**

This handbook applies to all crewed maintenance operations carried out on orbital platforms operated under a Northwind service contract. It does not cover launch or re-entry procedures, which are governed by the flight manual issued separately for each vehicle.

All personnel must complete the refresher assessment every 18 months. A crew member whose assessment has lapsed may not perform work outside the pressurised hull under any circumstances.

## **Chapter 2: Atmospheric Assessment**

Before any hatch is opened, the local atmosphere must be sampled and catalogued. Pressure, gas composition, ultraviolet radiation levels and surface gravity are all recorded in the station log before the assessment is considered complete.

If the atmosphere contains more than 2 per cent halogen gases, or if the pressure exceeds 3 terrestrial atmospheres, the environment is declared a Red Hazard Level, which obliges the crew to wear heavy pressure

suits of Class IV rating for the entire duration of the operation.

A Red Hazard Level declaration must be logged by two independent crew members. Automated declarations raised by the station monitoring system are advisory only and do not satisfy this requirement.

### **Chapter 3: Decompression Procedures**

Controlled decompression is performed in three stages, each lasting no less than 12 minutes. The intermediate stage may be extended if any crew member reports discomfort, and the extension does not need to be authorised.

### **Chapter 4: Quarantine and Biological Containment**

Any sample retrieved from an uncatalogued surface is treated as biologically active until proven otherwise. Samples travel in sealed Level 3 containment cases and are never opened aboard a crewed platform.

The final disembarkation at the orbital quarantine stations can last up to three months. During that period the crew remains under continuous medical observation and contact with the surface is limited to scheduled transmissions.

## **Chapter 5: First Contact Procedures**

In the event that an artificial structure of unknown origin is identified, no physical approach closer than 500 metres is permitted without authorisation from the mission director.

The observation window is fixed at 96 hours. During that window only passive sensors may be used; active scanning of any kind is expressly prohibited.

## **Chapter 6: Equipment Maintenance Intervals**

Pressure suits are inspected after every 40 hours of use. Seals are replaced every 12 months regardless of use. Helmet visors are replaced after any impact event, however minor.

## **Chapter 7: Emergency Communication**

The emergency beacon transmits on the reserved band for 72 hours on internal power. The beacon must be tested at the start of every rotation and the test recorded in the station log.